

The Mobility Census: Closed Forms and Maxima

Theorem monograph T03 of the chess-dynamics-lab program

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Keywords: mobility, closed form, maximum, census

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Setting

The mobility of a piece on the empty board — the number of squares it attacks — is the fundamental kinetic parameter of the particle. Mobility enters the position Lagrangian (T05) as the kinetic term and the search ordering heuristic (T10) as a weight. The task is to obtain closed mobility formulas for all pieces as functions of the square coordinates, to prove the maxima and the full sums, and to fix the argmax sets. All numbers are verified by protocol C3.

Theorem

Theorem T03 (the mobility census). On the empty board 8×8 the mobilities as functions of the square (f, r) , $0 \leq f, r \leq 7$, are: the rook — the constant 14; the bishop — $14 - |f - r| - |f + r - 7|$; the queen — the sum of the rook and bishop values; the king — 3, 5 or 8 (corners, edge without corners, interior); the knight — from 2 to 8 by the offset table. Maxima: king 8, knight 8, bishop 13, rook 14, queen 27; the argmax of the queen and the bishop is the central block $\{d4, e4, d5, e5\}$, the argmax of the rook is the whole board, the argmax sets of the king and the knight are their central zones. The full sums over the board: king 420, knight 336, bishop 560, rook 896, queen 1456.

Proof

The rook attacks every square of its file except its own (7 squares) and every square of its rank except its own (7 squares); the file and the rank intersect only at the own square, so there is no overlap and the mobility is $7 + 7 = 14$ from any square — a constant.

The bishop moves along the two diagonals through the square. The length of the $(+1, +1)$ diagonal through (f, r) equals $8 - |f - r|$: the difference $f - r$ is constant along it and runs from $-(7 - |f - r|)$ to $+(7 - |f - r|)$ — exactly $8 - |f - r|$ squares. Subtracting the square itself, we get $7 - |f - r|$ attacked squares on that diagonal. Likewise the anti-diagonal of direction $(+1, -1)$ has length $8 - |f + r - 7|$ and contributes $7 - |f + r - 7|$ attacked squares. The diagonals intersect only at the square itself, so the bishop mobility is $(7 - |f - r|) + (7 - |f + r - 7|) = 14 - |f - r| - |f + r - 7|$. The maximum is attained when $f = r \in \{3, 4\}$ and $f + r = 7$, i.e. on the squares where both $|f - r|$ and $|f + r - 7|$ are minimal: the central block $d4, e4, d5, e5$ gives $14 - 0 - 1 = 13$ (for $d4$: $|f - r| = 0$, $|f + r - 7| = 1$) — and this is the maximum, since both subtracted quantities are non-negative and their sum equals 1 on the whole central block.

The queen combines the rook lines and the bishop diagonals; their intersection is the own square only, so the queen formula is the sum $14 + (14 - |f - r| - |f + r - 7|) = 28 - |f - r| - |f + r - 7|$. The maximum 27 is attained on the same central block, and the argmax coincides with the bishop one. The king attacks up to 8 neighbouring squares: 3 in a corner, 5 on the edge without corners, 8 in the interior; the sum $12 + 120 + 288 = 420$ comes from the direct count of the squares of each type: 4 corners, 24 edge squares without corners, 36 interior squares.

The knight: eight offsets $(\pm 1, \pm 2), (\pm 2, \pm 1)$; the mobility equals 8 minus the number of offsets leaving the board. The full degree table (2 at the corners, 3 and 4 on the edge, 6 in the corner-diagonal ring, 8 in the centre) is verified by direct enumeration; its sum is 336 — consistent with T02 where $336/2 = 168$ edges of the knight graph. The maximum 8 is attained on the 16 central squares ($c3-f6$). All full sums: rook $64 \cdot 14 = 896$; bishop $\sum (14 - |f - r| - |f + r - 7|) = 896 - 168 - 168 = 560$; queen $896 + 560 = 1456$; king 420; knight 336. The theorem is proved.

Protocol data

The mobility census (check C3)

Piece	Maximum	Argmax	Board sum
King	8	26 interior squares	420
Knight	8	16 central squares	336
Bishop	13	$d4, e4, d5, e5$	560
Rook	14	all 64 squares	896
Queen	27	$d4, e4, d5, e5$	1456

Summary of what is proved

We obtained closed mobility formulas for all five pieces, proved the maxima 8/8/13/14/27 with the exact argmax sets and the full sums 420/336/560/896/1456. The mechanism is the geometry of diagonal lengths plus double counting; the agreement with the edge census of T02 is complete (the mobility sums equal the directed-move counts of the graphs). These constants are the kinetic weights of the whole

subsequent theory: the Lagrangian (T05), the threat fields (T04) and the search heuristic (T10).

Verification

Check: `python3 dynamics.py -run C3` or the C3 line of the polyglot core. The mobility tables are built by scanning all 64 squares for each piece, the closed forms and sums are compared with the computed ones; no discrepancies — PASS in all seven implementations.

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